

# Sahaj Patel

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## EDUCATION

### Georgia Institute of Technology

Atlanta, GA

*Bachelor of Science in Aerospace Engineering, Minor in Materials Science*

*August 2024 – May 2028 (Expected)*

- **GPA:** 4.00 / 4.00
- **Awards:** 2024 Stamps President's Scholar Semifinalist; Sigma Gamma Tau (AE Honors Society); 4x Faculty Honors; 4x Dean's List; Knack Tutor Excellence Award; Outstanding URA Presentation Award
- **Coursework:** Jet & Rocket Propulsion; AE Vehicle Performance; Control Systems; System Dynamics; Aerodynamics; Thermodynamics; Dynamics; Circuits; Differential Equations; Deformable Bodies; Statics; Engineering Materials; Engineering Graphics; Computing for Engineers; Manufacturing

## WORK HISTORY & EXPERIENCE

### Phoenix Air Group

Cartersville, GA

*Structural Design Engineering Intern*

*May 2026 – August 2026*

- Fabricated 100+ Multiple Antenna Processing Systems (MAPS) components for special-mission aircraft, supporting improved mission system integration and overall aircraft operational efficiency
- Developed 75+ CAD models and assembly layouts to analyze component fitment, identify design conflicts, and reduce prototype remake time by 30% during aircraft modification projects
- Analyzed 30+ aircraft components for corrosion damage and repair feasibility, reducing inspection turnaround time through standardized evaluation procedures

### Experimental Flights - Vertically Integrated Project @ Georgia Tech

Atlanta, GA

*Military Drones | Structures Engineer*

*August 2025 – Present*

- Reduced airframe weight by 18% by transitioning from 3D-printed PLA components to a carbon-fiber/foam composite structure, extending flight endurance from 12 to 18 minutes
- Executed 15+ test flights, identifying and resolving a loiter-stability control issue that improved successful mission completion rate from 60% to 90%

### Flowers Invention Studio @ Georgia Tech

Atlanta, GA

*Prototyping Instructuor / Manufacturing Engineer*

*August 2025 – Present*

- Skilled in utilizing 10+ fabrication tools including mills, lathes, waterjets, 3D printers, laser cutters, and woodworking machinery to design, prototype, and manufacture complex components
- Improved manufacturing processes, increasing efficiency by 20% and reducing material waste by 15% through optimized tooling and workflow strategies

## PROJECTS

### Bridge Structural Optimization | Statical Analysis

*August 2025 - December 2025*

- Analyzed force balance and internal member loads, identifying critical tension and compression members to guide structural optimization
- Evaluated additive manufacturing materials through load testing, achieving a maximum holding capacity of 125 lb

### Single-Propeller Cessna Airplane | Engineering Graphics & Design

*January 2025 – May 2025*

- Designed and assembled a 3-component aircraft system (fuselage, wing, propeller) into a single functional assembly using cantilever and axis-of-rotation snap-fit locking mechanisms
- Analyzed part tolerances and iteratively adjusted thicknesses and fillets, resulting in reliable assembly and repeatable manufacturing outcomes

### Aerodynamic Wooden Glider | Introduction to Aerospace Engineering

*August 2024 - December 2024*

- Engineered and fabricated a wooden glider using aerodynamic analysis, material selection, and adhesive optimization to maximize flight performance
- Achieved a 79 ft flight distance (2nd place) by iteratively refining angle of attack, aspect ratio, and center-of-mass placement through systematic testing

## SKILLS

**Technical:** Solidworks; MATLAB; ANSYS; R Studio; MagicDraw/Cameo; Microsoft, Google; Canva

**Operational:** Mill; Lathe; Waterjet; 3D Printer; Laser Cutter; Mechanical Tools

**Social:** Communication; Time Management; Organization; Proactivity; Critical Thinking; Problem Solving

**Languages:** English (fluent); Gujarati (semi-fluent); Spanish (conversational)